

Hawks are used by destroyers. More such helicopters are at times housed in the carriers also. These helicopters fly ahead of the CSG in a pattern and try to pin down the submarine. These have special sensors such as forward-looking infrared (FLIR) imaging systems, magnetic anomaly detectors (MADs), dunking sonars and dispensable sonobuoys to detect submarines.

FLIR imaging system. This is a high-power night-vision device used to detect surfaced submarines even in pitch darkness. The system senses the heat emitted by the submarines and is useful only when the enemy submarine has surfaced to get fresh air. In an FLIR, the surfaced submarine will stand out like a black spot on a plain,



Fig. 15: Hull-mounted sonar



Fig. 16: A torpedo being launched from an Aegis destroyer

white paper. But, what happens when the enemy submarine is sailing underwater? There comes MAD.

Magnetic anomaly detector. A submarine is a colossal mass of metal sailing underwater. Magnetic field disturbances are created by the metallic mass's interaction with the water's different temperature gradients. MAD picks up these disturbances and detects submarine activity. Then, it transmits the location of the submarine to the destroyer through datalinks.

MADs are only a part of the detection system. Whereas, to pin the exact location of the submarine, helicopter's dunking sonar and dispensable sonobuoys come into play.

Dunking sonar. The helicopter hovers over the water surface and dunks sonar equipment. After dunking, resulting sonar readings are recorded and transmitted to the destroyer. This, the helicopter does at various locations in the sea. In the destroyer, readings from sonars are processed in underwater warfare command-and-control (C²) systems. This system comes directly under Aegis C² system of the destroyer. Another method for detecting enemy submarine activity is through dispensable sonobuoys.

Dispensable sonobuoys. The helicopter strategically dispenses these sonobuoys at places where submarine activity is suspected or expected. These sonobuoys separate into two sections as sonar equipment and as a buoy. Sonars emit ultrasonic sound pulses as beeps and receive echoes like whales. Output of each beep is transmitted to the helicopter through buoys.

A sonar computer in the helicopter collects information from various sonobuoys and transmits data to the underwater warfare C² centre



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